



Trailer Battery & EcoFlow Monitoring Overview

Celestial Electric Solar & Battery Room Monitor

Rev 2 - Trailer Battery and EcoFlow Monitoring Update

SOLAR & BATTERY ROOM MONITOR

LOW-VOLTAGE MONITORING. REAL-TIME VISIBILITY. PEACE OF MIND.

- TEMPERATURE MONITORING
- HUMIDITY MONITORING
- WATER LEAK DETECTION
- DOOR / CONTACT STATUS
- CLOUD DASHBOARD & ALERTS

TEMPERATURE
72.4 °F

HUMIDITY
45 %

WATER LEAK
DRY
No water detected

DOOR / CONTACT
CLOSED
Door is closed

STATUS
All systems normal
Last updated: 11:32 AM

LOW-VOLTAGE MONITORING ONLY

ARDUINO CLOUD CONNECTED

REAL-TIME ALERTS & NOTIFICATIONS

BUILD BETTER. MONITOR SMARTER.

Prepared for Arduino Project Hub public documentation and downloadable project files.

Purpose

This document adds trailer-specific battery and mobile power information to the Celestial Electric Solar & Battery Room Monitor project. The project is intended to demonstrate low-voltage monitoring for a mobile power/service trailer, including battery compartment conditions, basic 12V bank visibility, water/leak detection, access/contact status, and dashboard reporting.

Monitored Battery System

Item	Project Detail
Primary monitored 12V bank	Two Duracell 12V lead-acid batteries, each rated 810 cold cranking amps (CCA), installed in parallel.
Parallel wiring meaning	The batteries remain a 12V nominal battery bank. Available starting current and reserve capacity are increased when the batteries are properly matched and wired.
Nominal combined CCA	Two matched 810 CCA batteries in parallel are nominally capable of about 1,620 CCA. This is a starting-current rating, not a usable energy-capacity rating.
Monitoring target	Bank voltage, compartment temperature/humidity, water/leak status, door/contact status, local normal/alert LED indication, and cloud dashboard visibility.
EcoFlow system	The trailer also includes an EcoFlow portable power system. The exact EcoFlow model, capacity, and solar/AC input ratings should be confirmed from the unit nameplate and added to the final documentation.
Trailer role	Celestial Electric mobile service, demonstration, and training trailer used to show low-voltage monitoring, field diagnostics, and mobile energy concepts.

Important battery note: cold cranking amps (CCA) is a starting-current rating. It does not equal amp-hours, watt-hours, runtime, or available stored energy. For runtime estimates, use the actual amp-hour or reserve-capacity rating from the battery labels or manufacturer data.

Trailer Monitoring Scope

The Arduino project can be presented as a monitoring and training layer for the trailer, not as the trailer power controller. The first public version should focus on visibility and safety boundaries.

- Monitor the 12V Duracell battery bank voltage through a fused, low-current sensing circuit.
- Monitor battery/power compartment temperature and humidity.
- Monitor water/leak detection near the trailer power equipment area.
- Monitor compartment door or access-panel contact status.
- Show local normal/alert LEDs for quick visual status.
- Send status values to an Arduino Cloud dashboard for demonstration and training.
- Document the EcoFlow portable power system as a separate mobile power source; add exact model details after confirming the nameplate.

Recommended Dashboard Fields

Dashboard Variable	Purpose
batteryBankVoltage	Displays the approximate 12V Duracell battery bank voltage using a fused voltage sense input.
batteryCompartmentTemp	Tracks compartment temperature near the battery or power equipment area.
batteryCompartmentHumidity	Tracks humidity in the monitored compartment.
waterLeakDetected	Shows whether water is detected near the equipment compartment or trailer power area.
doorContactStatus	Shows whether the compartment door, access door, or enclosure contact is open or closed.
ecoFlowStatus	Optional manual or future integration field for EcoFlow system status.
systemStatus	Simple summary: Normal, Attention Required, or Alert.

System Boundary and Safety

Area	Boundary
Arduino controller	Monitoring-only demonstration device. It does not start, stop, charge, discharge, transfer, isolate, or protect any battery, generator, inverter, EcoFlow unit, or electrical system.
12V battery sensing	Use fused low-current sense leads and proper voltage scaling or isolation. Do not route starter current, inverter current, charger current, or trailer load current through the Arduino.
EcoFlow system	Monitor status/environment only unless using manufacturer-approved APIs, ports, and accessories. Do not modify internal EcoFlow circuitry.
Field installation	Use listed equipment, proper fusing, strain relief, enclosures, wire separation, and code-compliant wiring methods.

Suggested Trailer Description for Project Hub

The monitored trailer is a Celestial Electric mobile service, demonstration, and training platform. The trailer power area includes a 12V auxiliary battery bank made from two Duracell 12V lead-acid batteries rated 810 CCA each, wired in parallel for a 12V nominal bank. The trailer also includes an EcoFlow portable power system used as a mobile power resource. This Arduino project demonstrates how low-voltage sensors can provide visibility into the trailer power compartment, environmental conditions, water/leak status, and access/contact status without controlling or replacing the actual battery, inverter, charging, generator, or safety equipment.

Information Still To Confirm

- Exact Duracell battery model number and group size from the label.
- Amp-hour or reserve-capacity rating, if shown on the batteries.
- EcoFlow model name and rated capacity in watt-hours.
- EcoFlow AC input, solar input, and maximum output ratings from the nameplate or user manual.
- Trailer battery charging source: vehicle alternator, shore charger, solar charger, DC-DC charger, or other.
- Whether the Arduino will monitor voltage only, or voltage plus current using a proper listed/shunted battery monitor interface.

Public Project Disclaimer

This project is for low-voltage monitoring, training, and demonstration only. It is not a replacement for listed safety controls, battery management systems, manufacturer monitoring, generator/ATS controls, utility-required protection, fire alarm systems, or code-required protective devices. Any real trailer installation must use proper fusing, wiring, enclosures, strain relief, isolation, and code-compliant methods.